

Breaking through Accessed Bit Limits of DAMON

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LINUX
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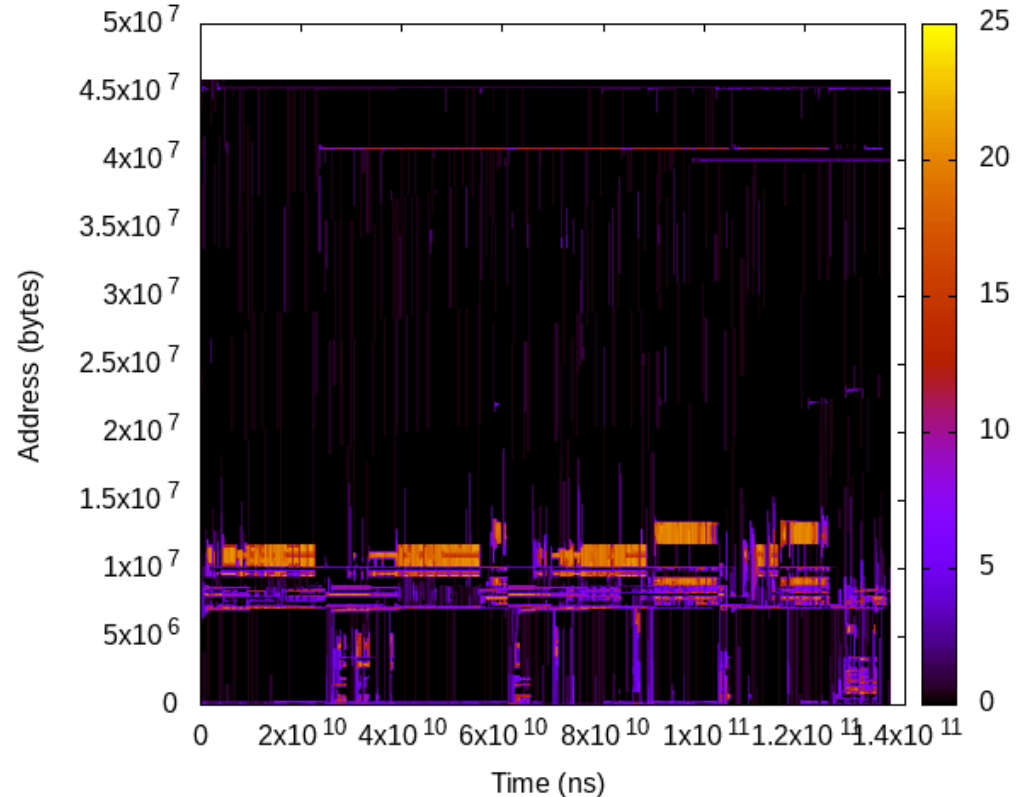


DAMON in One Slide

- Spawn a kernel thread that does
- See if each page is accessed, every 5ms
- Inform users the findings, every 100ms
- Time intervals can be [auto-]tuned

Providing Access Information

- Location
- Frequency
- Stability
- Recency



Access pattern heatmap of
Splash2x/Raytrace

Page Table Accessed Bit (A-bit)

- DAMON's access check primitive
- H/W sets the bit when it access the page
- S/W (DAMON) can clear the bit

A-bit Limits

- Lack of info
 - No accessor (which CPU/proc/cgroup)
 - No access type (read/write)
- Page table (reverse) walk overhead
- (Huge) Page granularity

Idea to Break-through Limits

- Extend DAMON for Data Attributes
- Attribute Examples
 - Memory is backed by cgroup X
 - A-bit of the data is set
 - AMD IBS found access to the data
(with accessor and access type info)

Roadmap

- Extend DAMON for Data Attributes
 - Milestone 1, by LPC'26
- Support perf event based access attribute
 - Milestone 2, by LSFMMBPF'27
- Support AMD IBS, Intel PEBS, ARM SPE, ...

Milestone 1: Complete

- DAMON is (almost) renamed to DAMON
 - Data Access/Attribute Monitoring and Operations eNgin
- DAMOS update is in progress

Milestone 2: In Progress

- Perf cycle event based access monitoring
 - Akinobu suggested initial form
 - Ravi suggested cross-vendor perf substrate
 - SJ is revisioning
- Decode perf sample load/store instruction

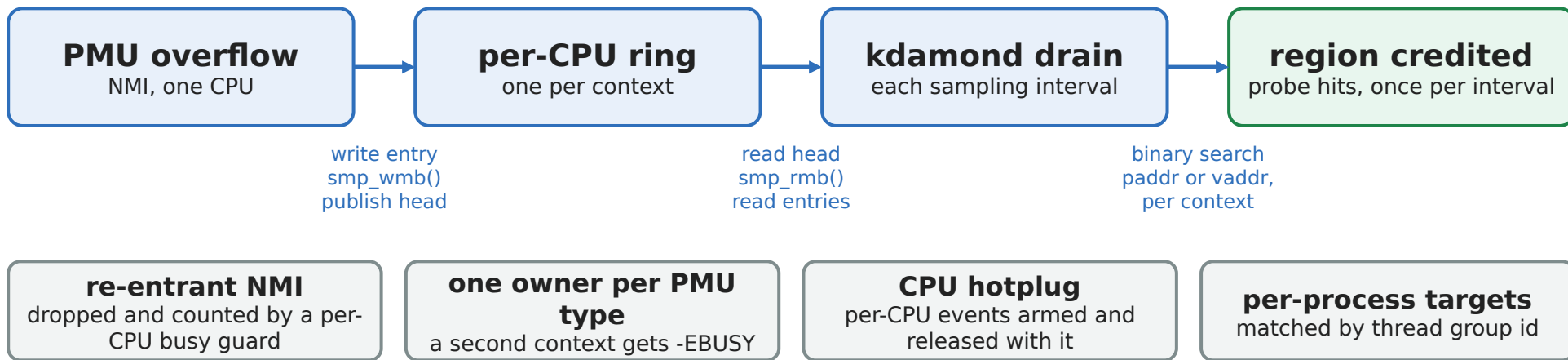
Milestone 3: Prototypes Ready

- Ravi: AMD IBS, Intel PEBS Support
- Kunwu: Arm SPE Support

Milestones 2&3, More Details

Milestone 2: NMI-Safe Delivery

PMU overflow runs in NMI; `damon_report_access()` takes a mutex, so delivery is deferred



Next, with Akinobu: fold into milestone 2's report path · `damon_report_access()` made lock-free

Milestone 3: IBS, PEBS, SPE

TODAY · ACCESSED-BIT SCAN

page-table walk
reads and clears the Accessed bit

one look per sampling interval
accessed or not, no more

THIS WORK · PERF EVENTS

Intel PEBS
virtual address

AMD IBS
physical address

Arm SPE
Kunwu Chan's backend

perf overflow handler · NMI context
turns each sample into an access report

AUX trace buffer
kdamond parses it, no NMI

per-context, per-CPU ring
lockless, NMI-safe

kdamond drain
each sampling interval

access rate (scan) · per-probe hits (perf, once per interval)

access pattern + probe_hits_wsum → DAMOS: promote · demote

Probe weight 0: the scan sets the access rate; perf samples are probe hits. Weights set: scan off, weighted hits decide.

Built on [Akinobu Mita's perf-event access check](#) and co-developed with him · [RFC v4, October 2026](#)

QnA

Beyond Weighted Interleaving: Bandwidth-Driven Memory Tiering with DAMON

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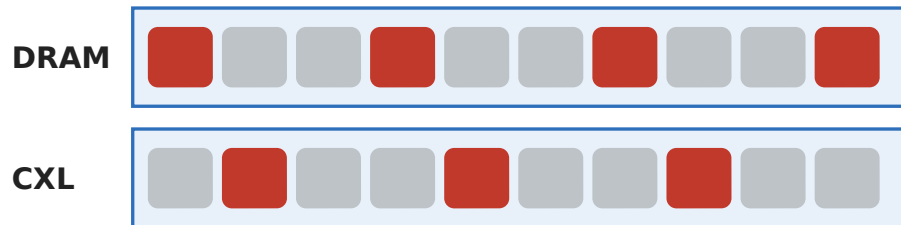
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Each Policy Solves Half

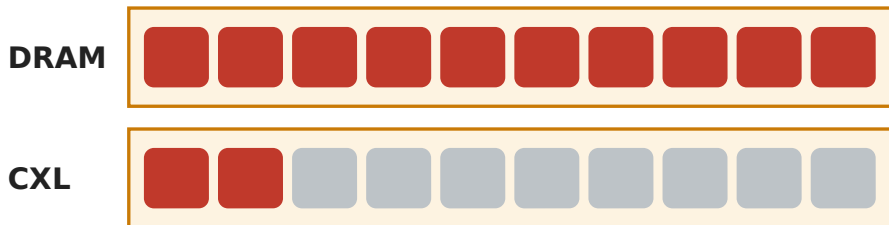
WEIGHTED INTERLEAVING · fixed split



✓ drives both memory paths at once

✗ blind to which pages are hot

HOTNESS TIERING · promote what is hot

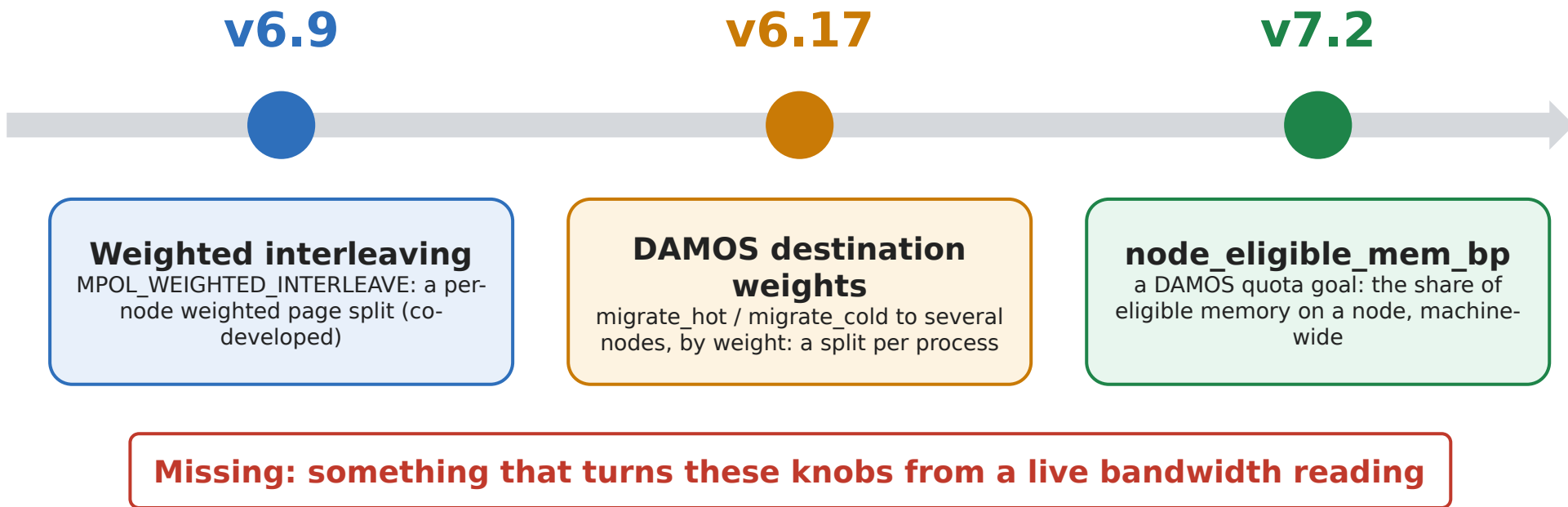


✓ right about heat

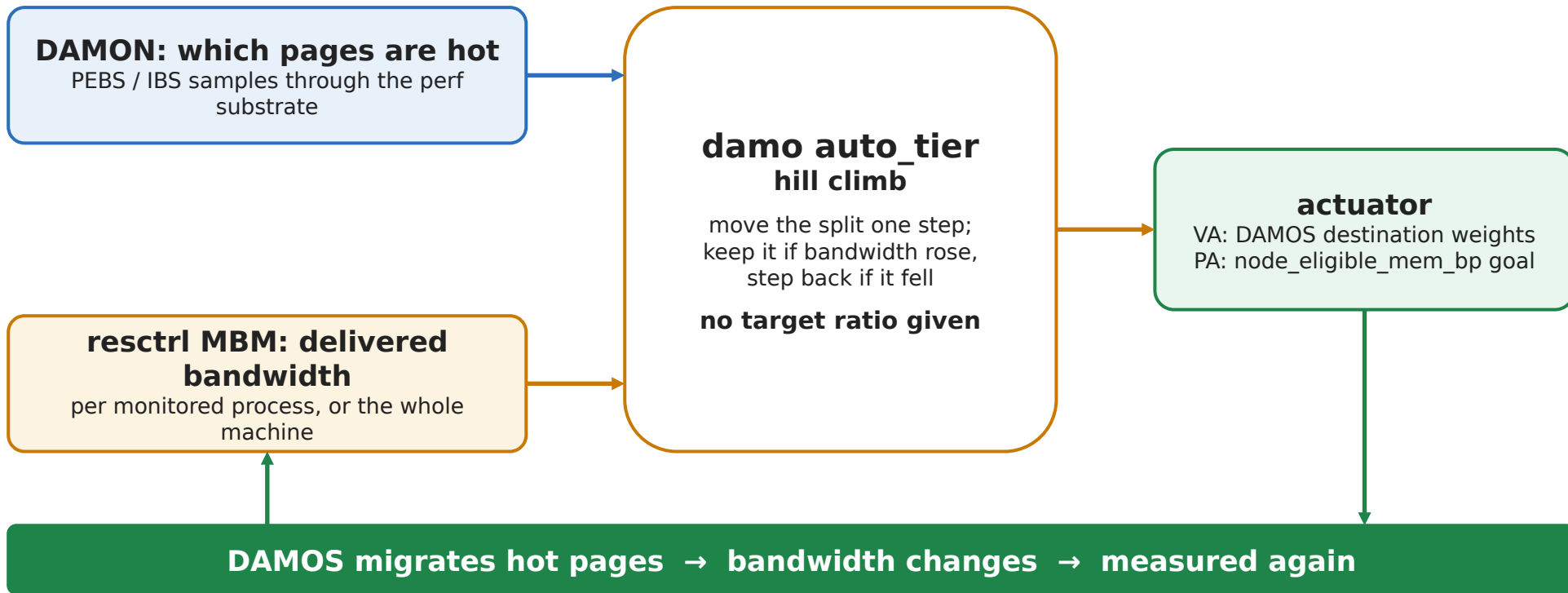
✗ keeps filling DRAM after it stops helping

Wanted: place by hotness, split by the bandwidth the machine actually delivers

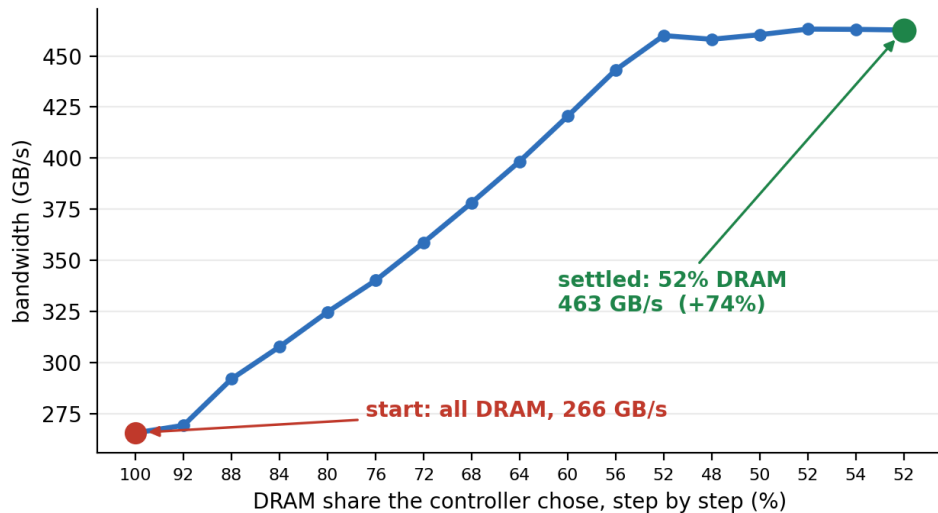
The Knobs Are Already Upstream



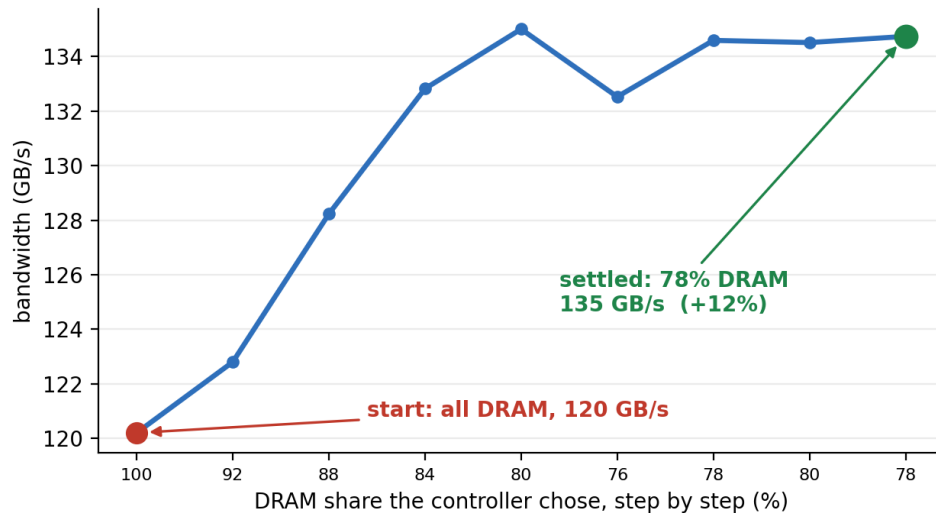
Closing the Loop



It Finds the Best Split on Its Own



Intel PEBS · virtual addresses · named processes
266 → 463 GB/s, settled at 52% DRAM



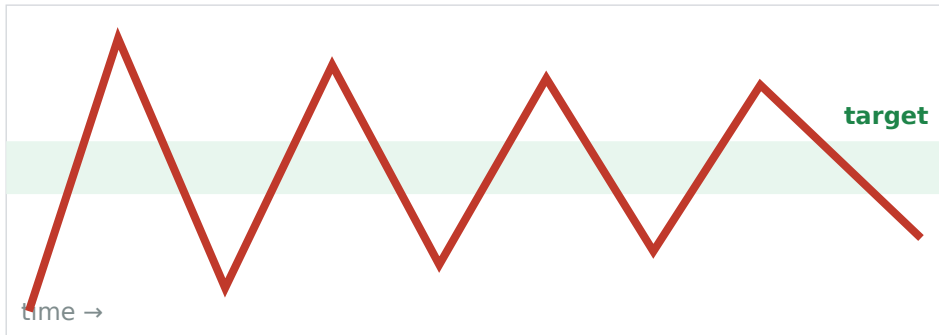
AMD IBS · physical addresses · whole machine
120 → 135 GB/s, settled at 78% DRAM

Both start all-DRAM, with no target and no model of the machine

Why Hardware Samples Settle

ACCESSED-BIT SCAN · a delayed view

pages move, but the scan reports it
several aggregation intervals later
so the controller corrects an error already fixed



✗ overshoots, then hunts around the target

Measured, accessed-bit scan

even with a cache workaround: in band only 15-22% of the time

HARDWARE SAMPLES · a current view

the next samples land where
the pages are now
so the goal is measured against real placement



✓ reaches the target and holds

Measured, IBS samples

the climb reached its peak in 4 steps (previous slide)

Traces are illustrative.

What Next: for Discussion

Upstream path

RFC v4 on damon/next; fold into milestone 2's report path with Akinobu

Where the controller lives

userspace damo today; which parts belong in DAMOS goal tuning?

More hotness sources

sampling PMUs feed the overflow path; trace PMUs such as Arm SPE via Kunwu's AUX backend

Observability

damos_node_eligible_mem_bp tracepoint, posted for mm-new

Kernel: [RFC v4 cover letter](#) · [tracepoint patch](#)

Userspace: [damo_auto_tier branch](#)